Alignment-Free phylum identification of COI sequences: An approach based on K-mers, entropy profiling and deep learning

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Processing

A)

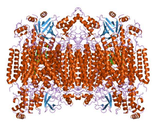




Figure A. Cytochrome c Oxidase subunit I (COI) as a molecular marker. This mitochondrial transmembrane enzyme is fundamental to cellular respiration. Due to its accelerated mutation rate, which facilitates species differentiation, and the presence of conserved flanking regions, the gene encoding COI has been established as the global standard for DNA barcoding in animals.

B)

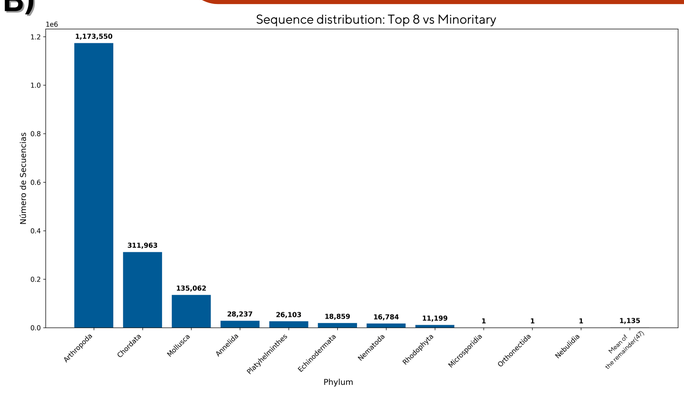
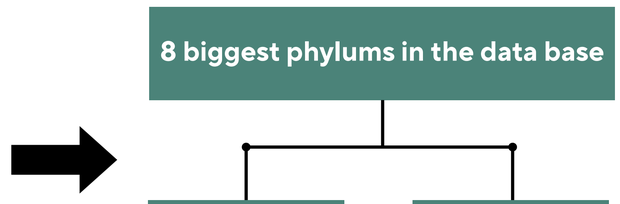


Figure B. Distribution of COI gene sequences by Phylum in the MIDORI2 database. The chart illustrates the disparity in biological data availability. The blue bars represent the 8 most abundant Phyla, selected for model training to ensure statistical robustness. The red bars show representative examples of minority groups, while the gray bar indicates the average sequence count for the remaining taxa ("Remainder"). A long-tail distribution is evident, justifying the exclusion of underrepresented groups to prevent bias in machine learning tasks.



1 Shannon Entropy-based Subsequence Selectionce search

A)

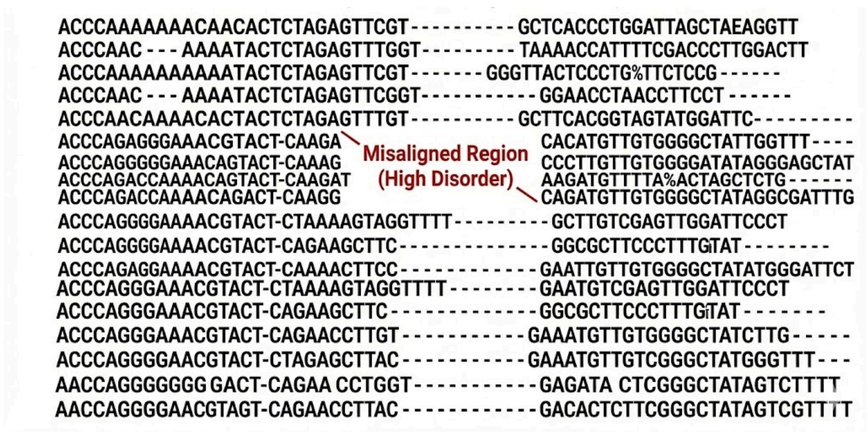
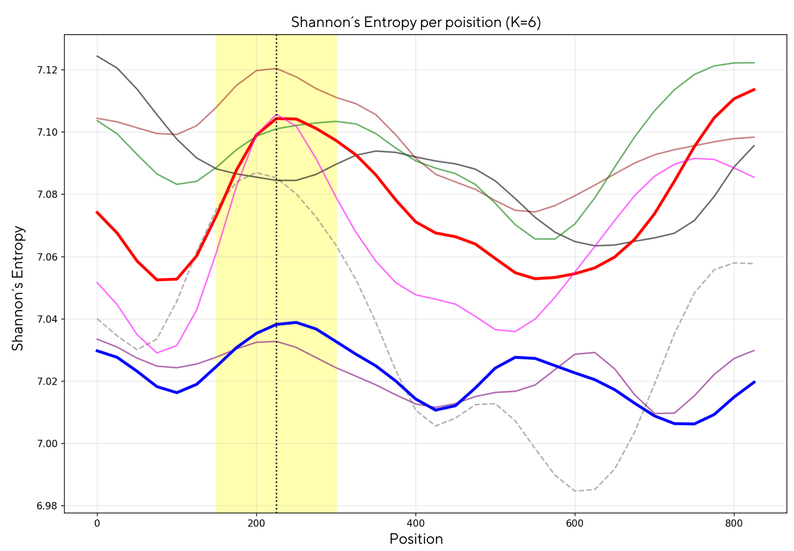


Figure A. Multiple Sequence Alignment (MSA) of CO1 sequences. The high variability and differing lengths of these sequences resulted in a highly gapped alignment. These characteristics rendered whole-sequence analysis computationally challenging when processing the complete database of 1.7 million sequences.

B)



$$H = -\sum_{i=1}^{k} p_i * \log_2(p_i)$$

Figure B. Identification of the Optimal Classification Subsequence. CO1 gene variability profile calculated using Shannon Entropy on 6-mers (K=6). The curves illustrate the information distribution across different phyla. The yellow shaded area highlights a 150 bp window centered around the maximum entropy peak. This region was selected for maximizing variability within each phylum, capturing the most discriminative information for species identification.

2 Genomic Representation via FCGR and k-mer Frequencies

A)

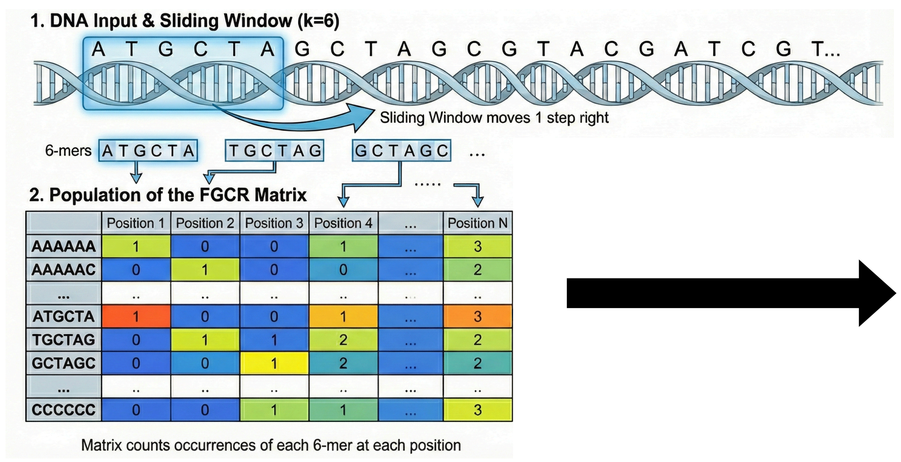


Figure A. Generation of the Frequency Chaos Game Representation (FCGR) matrix. After selecting the optimal window and subsequence, the matrix is generated by computing the frequency of each k-mer. These counts are then mapped to specific grid coordinates, resulting in a density representation of the sequence.

B)

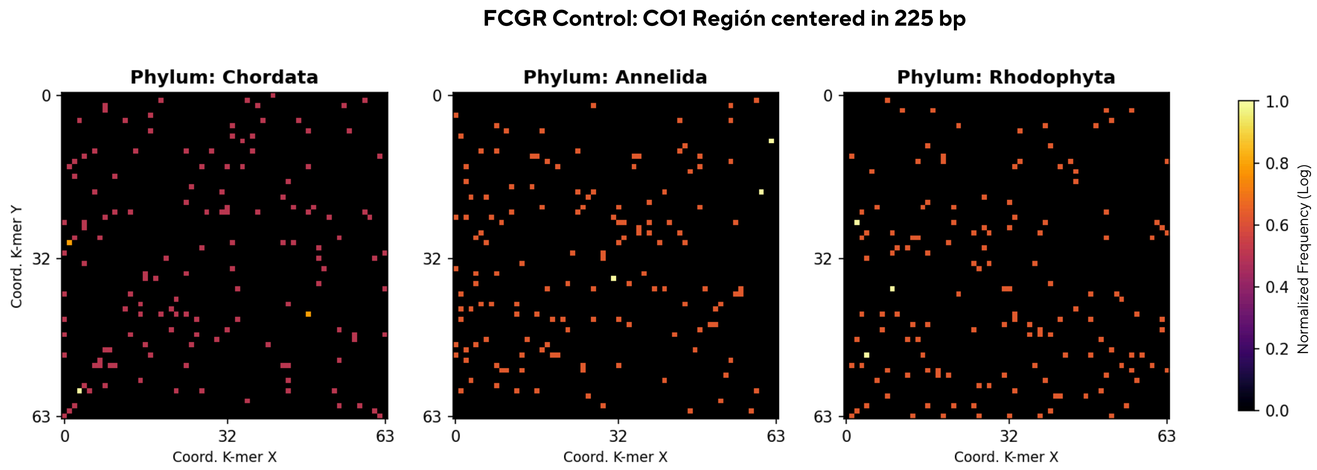
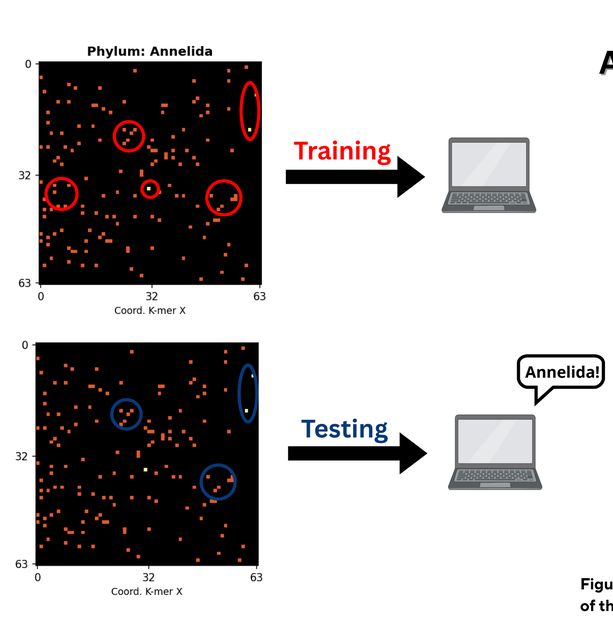


Figure B. Genomic Fingerprints: FCGR Representation (K=6). Transformation of the optimal subsequence into 2D image space for three distinct phyla (Chordata, Annelida, Rhodophyta). Each pixel in the matrix corresponds to a specific k-mer. Color intensity indicates the most frequent k-mers in that sequence. Note how each taxonomic group generates a unique texture pattern, enabling automatic classification via Convolutional Neural Networks (CNN).

3 Taxonomic Classification and Performance Evaluation



A)

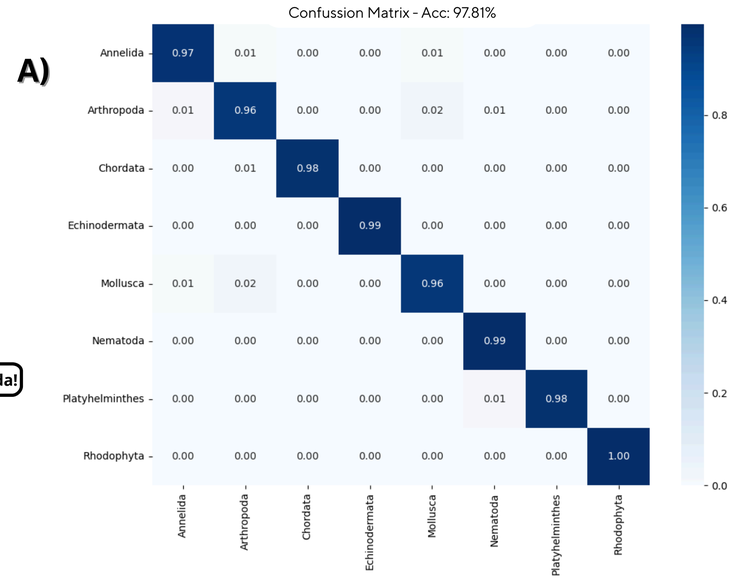


Figure A. Performance Evaluation on the Testing Set. Normalized confusion matrix applied to unseen data (20% of the dataset) never seen by the model during training. The neural network achieved a global Accuracy of 97.81%, demonstrating high generalization capability. The strong dark blue diagonal confirms that the FCGR representation effectively distinguishes taxonomic patterns across the 8 analyzed phyla with minimal classification errors.

B)

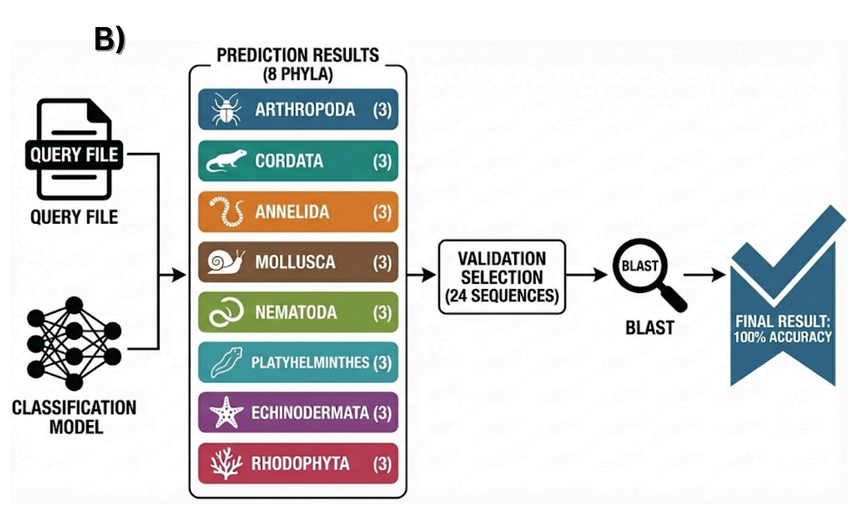


Figure B. Workflow for the external validation of the taxonomic classification model. Unknown query sequences were processed by the classification model, generating predictions across eight distinct phyla. To validate these predictions, a subset of 24 representative sequences (three from each predicted phylum) was selected. These sequences were independently identified using BLAST, confirming a 100% accuracy rate for the model's predictions in this validation set.

Conclusions & Future Perspectives

Our model demonstrated robust performance by confidently identifying COI gene sequences across 8 distinct phyla, validating the efficacy of combining an alignment-free approach based on k-mer frequencies with Deep Learning. This methodology proved to be highly scalable. As data availability increases, the framework can be readily extended to include additional phyla beyond the scope of this study. Furthermore, the FCGR representation of these k-mer distributions captures sequence patterns rich enough to enable future classification at finer taxonomic levels, such as Class or Order.

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